

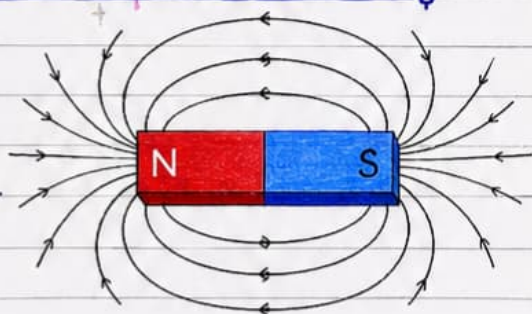


Chapter - 5

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Magnetism and Matter



Magnetic field lines

1. INTRODUCTION

- Magnetism is a natural phenomenon by which certain materials experience a force of attraction or repulsion.
- A magnet produces a **magnetic field** around it.

2. MAGNETIC FIELD ($\vec{B}$)

- Magnetic field is the region around a magnet or current carrying conductor, where a magnetic pole experiences a force.
- SI Unit : **Tesla (T)**

$$* 1 \text{ Tesla} = 1 \text{ N/A}\cdot\text{m}$$

$$* \text{ Dimensional formula: } [M^0 L^0 T^{-2} A^{-1}]$$

3. MAGNETIC FIELD LINES

- Magnetic field lines are imaginary closed curves.
- Outside a magnet, they emerge from **North pole** and enter into **South pole**.
- Inside a magnet, they go from **South pole** to **North pole**.
- No two field lines intersect each other.
- Closer the field lines, **stronger** is the field.

Properties

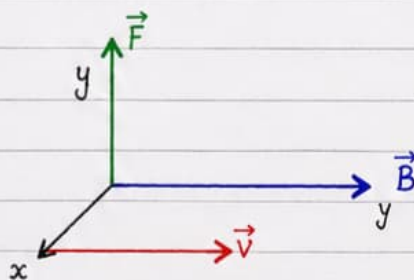
- They never intersect.
- They form closed loops.
- Direction is from N to S outside the magnet.
- Density indicates strength of field.

4. FORCE ON A MOVING CHARGE IN A MAGNETIC FIELD

- When a charged particle moves in a magnetic field, it experiences a force.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Where, q = charge, $\vec{v}$ = velocity, $\vec{B}$ = magnetic field

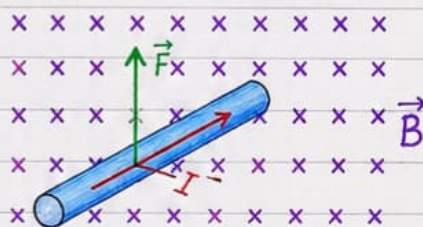


5. FORCE ON A CURRENT CARRYING CONDUCTOR

- When a current carrying conductor is placed in a magnetic field, it experiences a force.

$$\vec{F} = I(\vec{L} \times \vec{B})$$

Where, I = current, $\vec{L}$ = length vector in direction of current



Remember

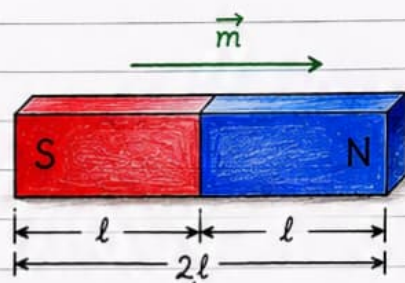
- Right Hand Thumb Rule** → For direction of Force.
- Fleming's Left Hand Rule** → For direction of Force on a current carrying conductor.



6. MAGNETIC MOMENT ($\vec{m}$)

- A bar magnet has two equal and opposite poles.
- Magnetic moment is a vector quantity.
- It is directed from South pole to North pole outside the magnet.
- SI Unit : Ampere metre² ($A \cdot m^2$)
- For a bar magnet : $m = M \times 2l$

Where, M = pole strength, $2l$ = length of magnet



Key Point

Magnetic moment tries to align itself in the direction of external magnetic field.



7. TORQUE ON A MAGNETIC DIPOLE

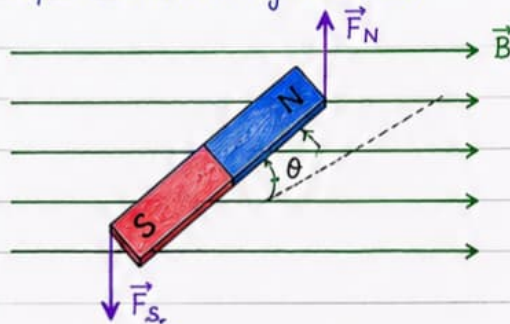
- When a magnetic dipole is placed in a uniform magnetic field, it experiences a torque.
- The dipole tends to align itself parallel to $\vec{B}$.
- Magnitude of torque,

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$$|\tau| = mB \sin \theta$$

Where, θ = angle between $\vec{m}$ and $\vec{B}$

Dipole in a uniform magnetic field $\vec{B}$



8. WORK DONE TO ROTATE A DIPOLE

- Work done to rotate a dipole from angle θ_1 to θ_2 in a uniform magnetic field B ,

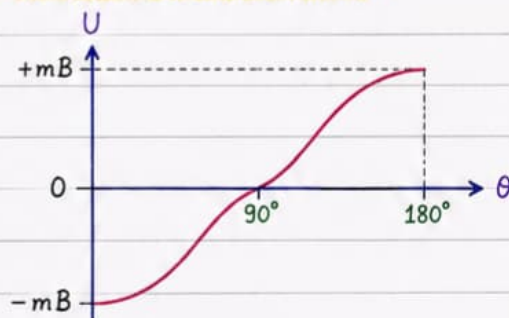
$$W = mB (\cos \theta_1 - \cos \theta_2)$$

9. POTENTIAL ENERGY OF A DIPOLE

- Potential energy of a dipole in a magnetic field at angle θ is,

$$U = -\vec{m} \cdot \vec{B} = -mB \cos \theta$$

ENERGY vs ANGLE GRAPH



Important Cases ★

- $\theta = 0^\circ \Rightarrow U = -mB$ (Minimum)
- $\theta = 90^\circ \Rightarrow U = 0$
- $\theta = 180^\circ \Rightarrow U = +mB$ (Maximum)

10. MAGNETIC MATERIALS

Type	Meaning	Examples	Key Properties
Ferromagnetic	Strongly attracted by a magnet and can be magnetised permanently.	Iron (Fe), Cobalt (Co), Nickel (Ni), Gadolinium (Gd)	- Very high permeability - Large positive susceptibility - Retain magnetism
Paramagnetic	Weakly attracted by a magnet. Magnetisation is temporary.	Aluminium (Al), Platinum (Pt), Magnesium (Mg), Oxygen (O ₂)	- Small positive susceptibility - Low permeability.
Diamagnetic	Weakly repelled by a magnet. Never gets magnetised.	Bismuth (Bi), Copper (Cu), Silver (Ag), Water, Mercury	- Negative susceptibility - Very low permeability

Quick Notes

- All materials possess diamagnetic property.
- Ferromagnetic materials are used to make permanent magnets.
- Soft iron is used in electromagnets.

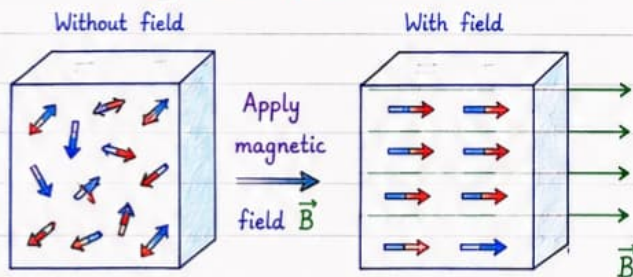


11. MAGNETISATION ($\vec{M}$)

- When a magnetic material is placed in an external magnetic field, each microscopic magnetic dipole aligns in a particular direction. The net magnetic dipole moment per unit volume is called **magnetisation**.
- SI Unit : Amperre per metre (A/m)

$$\vec{M} = \frac{\text{Net magnetic dipole moment } (\vec{m})}{\text{Volume } (V)}$$

Alignment of dipoles in a material



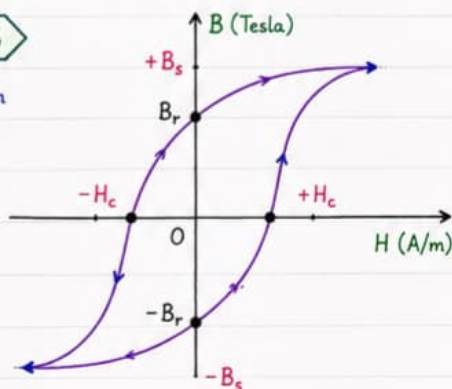
More alignment $\Rightarrow$ More magnetisation

12. BEHAVIOUR OF MATERIALS IN A MAGNETIC FIELD

Type of Material	Effect	Magnetic Susceptibility (χ)	Relative Permeability (μ_r)	Examples	Diagram
Diamagnetic	Weakly repelled by magnetic field.	$\chi < 0$ (Very small negative)	μ_r slightly less than 1	Bismuth, Copper, Silver, Water, Mercury	
Paramagnetic	Weakly attracted by magnetic field.	$\chi > 0$ (Very small positive)	μ_r slightly more than 1	Aluminium, Platinum, Magnesium, Oxygen, Sodium	
Ferromagnetic	Strongly attracted by magnetic field. Can be magnetised permanently.	$\chi \gg 0$ (Large positive)	$\mu_r \gg 1$ (Very large)	Iron, Cobalt, Nickel, Gadolinium	

13. HYSTERESIS IN FERROMAGNETIC MATERIALS

- When a ferromagnetic material is magnetised and then demagnetised, its magnetisation does not become zero even when the applied field becomes zero.
- The graph between B (magnetic flux density) and H (magnetising field) is called **hysteresis loop**.
- Retentivity (B_r)** : Residual magnetism when $H = 0$.
- Coercivity (H_c)** : Reverse field required to make $B = 0$.

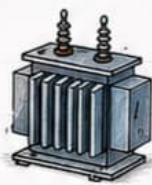


B_s = Saturation flux density
 B_r = Retentivity
 H_c = Coercivity
 Area of loop = Energy loss per cycle (hysteresis loss)

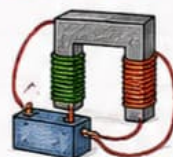
14. APPLICATIONS

- Soft iron (low retentivity, low coercivity)
 $\rightarrow$ Used in transformers, electromagnets, relays.
- Hard steel (high retentivity, high coercivity)
 $\rightarrow$ Used in permanent magnets, compass needles, loudspeakers.
- Magnetic materials are used in data storage, electric motors, generators, magnetic shielding.

Examples



Transformer



Electromagnet



Loudspeaker



Data Storage

QUICK SUMMARY

- Magnetisation ($\vec{M}$) $\Rightarrow$ dipole moment per unit volume.
- Diamagnetic $\Rightarrow$ repelled ($\chi < 0$)
- Paramagnetic $\Rightarrow$ weakly attracted ($\chi > 0$)
- Ferromagnetic $\Rightarrow$ strongly attracted ($\chi \gg 0$)
- Hysteresis loop shows retentivity and coercivity.

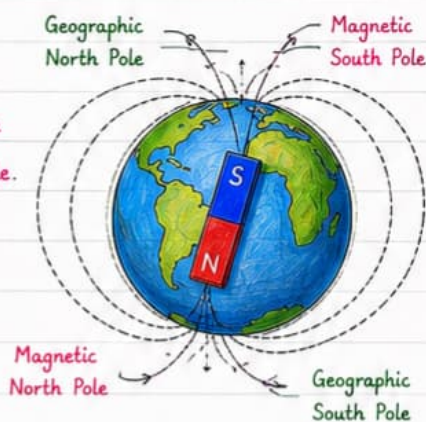
KEY TAKEAWAY

Magnetism in matter arises due to alignment of tiny magnetic dipoles. The response of a material depends on how easily these dipoles align with an external magnetic field.



15. EARTH AS A MAGNET

- The Earth behaves like a giant bar magnet.
- Geographic North pole lies near **Magnetic South pole** and Geographic South pole lies near **Magnetic North pole**.
- A freely suspended magnetic needle aligns itself in North-South direction.



Note

Declination is the angle between magnetic north and true north.

Inclination is the angle made by the magnetic field with horizontal.

16. SUMMARY TABLE

Concept	Formula / Relation	SI Unit	Meaning
Magnetic Field (B)	$B = \frac{F}{qv} \quad (v \perp B)$	Tesla (T)	Strength of magnetic field.
Magnetic Moment (m)	$m = I A \hat{n}$	$A \cdot m^2$	Vector representing strength and direction of a magnetic dipole.
Torque on Dipole (τ)	$\vec{\tau} = m B \sin \theta$	$N \cdot m$	Turning effect experienced by a dipole in magnetic field.
Potential Energy (U)	$U = -m B \cos \theta$	Joule (J)	Energy of dipole in magnetic field.
Magnetisation (M)	$M = \frac{\text{Net } \vec{m}}{V}$	A/m	Magnetic dipole moment per unit volume.
Magnetic Susceptibility (χ)	$\chi = \frac{M}{H}$	Dimensionless	Measure of how much a material can be magnetised.

17. IMPORTANT POINTS

- Magnetic field lines never intersect.
- Inside a magnet, field lines go from South to North.
- Magnetic materials improve magnetic field lines.
- Diamagnetic materials are weakly repelled ($\chi < 0$).
- Paramagnetic materials are weakly attracted ($\chi > 0$).
- Ferromagnetic materials are strongly attracted ($\chi \gg 0$).
- Hysteresis shows energy loss in a cycle.

18. NUMERICAL EXAMPLE

Q. A magnetic dipole of moment $2.0 \text{ A} \cdot \text{m}^2$ is placed in a uniform magnetic field of 0.50 T . If the dipole is aligned at 30° with the field, find (i) torque on it (ii) potential energy of the dipole.

Given: $m = 2.0 \text{ A} \cdot \text{m}^2$, $B = 0.50 \text{ T}$, $\theta = 30^\circ$

Solution: (i) $\vec{\tau} = m B \sin \theta$

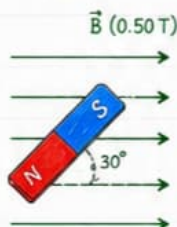
$$= 2.0 \times 0.50 \times \sin 30^\circ$$

$$= 2.0 \times 0.50 \times \frac{1}{2} = 0.50 \text{ N} \cdot \text{m}$$

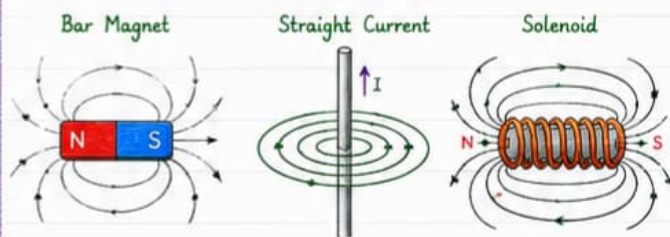
(ii) $U = -m B \cos \theta$

$$= -2.0 \times 0.50 \times \cos 30^\circ$$

$$= -1.0 \times 0.866 = -0.866 \text{ J}$$



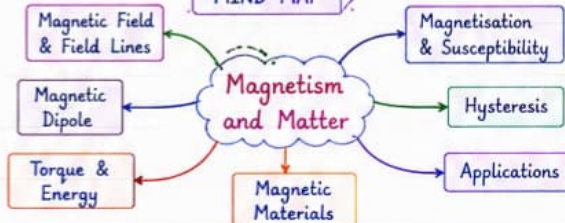
FIELD LINES: QUICK VIEW



QUICK FACTS

- Magnetism is a fundamental force.
- Only moving charges produce magnetic field.
- Earth's magnetic field protects us from solar winds and cosmic rays.
- Soft iron is used in electromagnets because it is easily magnetised and demagnetised.

MIND MAP



SELF CHECK

- Define magnetic moment.
- Write the formula for torque on a magnetic dipole.
- What is the SI unit of magnetic field?
- Classify materials into diamagnetic, paramagnetic and ferromagnetic with one example each.

KEY TAKEAWAY

Magnetism explains how materials interact with magnetic fields. Understanding magnetic dipoles, their behaviour in fields, and properties of materials is essential for many natural phenomena and technological applications.

